

# Matteo Palmonari

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## 5.3

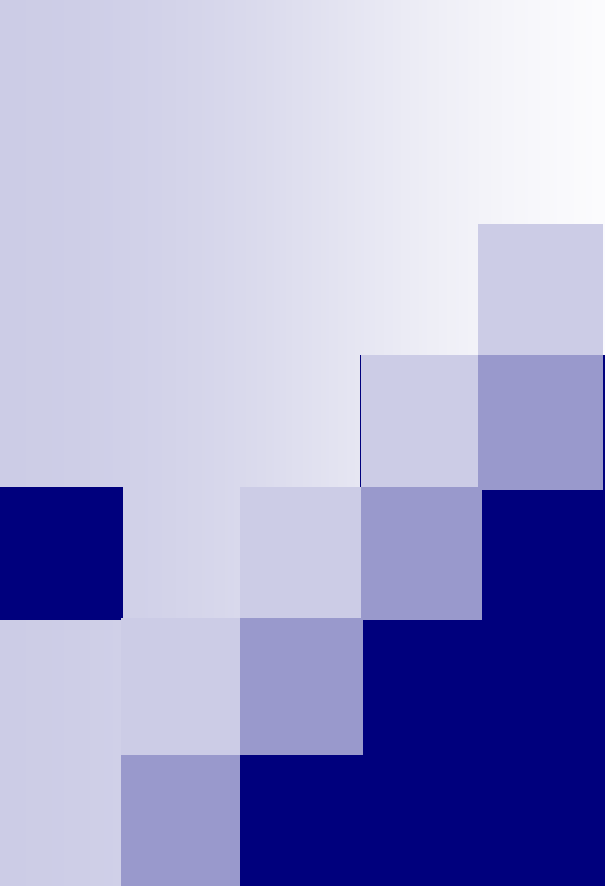
### *Information Extraction #2*



dat·AI data \* AI  
AI \* data

# Outline

- Some preliminaries about linguistic analysis
- Named Entity Recognition
- **Named Entity Linking**
- Relation Extraction



# Named Entity Linking

## Overview and Examples

# NEL - Surveys

## ■ Well-formed texts

- Shen, Wei, Jianyong Wang, and Jiawei Han. **Entity linking with a knowledge base: Issues, techniques, and solutions.** IEEE Transactions on Knowledge and Data Engineering 27.2 (2015): 443-460.

## ■ Tweets

- Derczynski, L., Maynard, D., Rizzo, G., van Erp, M., Gorrell, G., Troncy, R., ... & Bontcheva, K. (2015). **Analysis of named entity recognition and linking for tweets.** Information Processing & Management, 51(2), 32-49.

# NEL systems

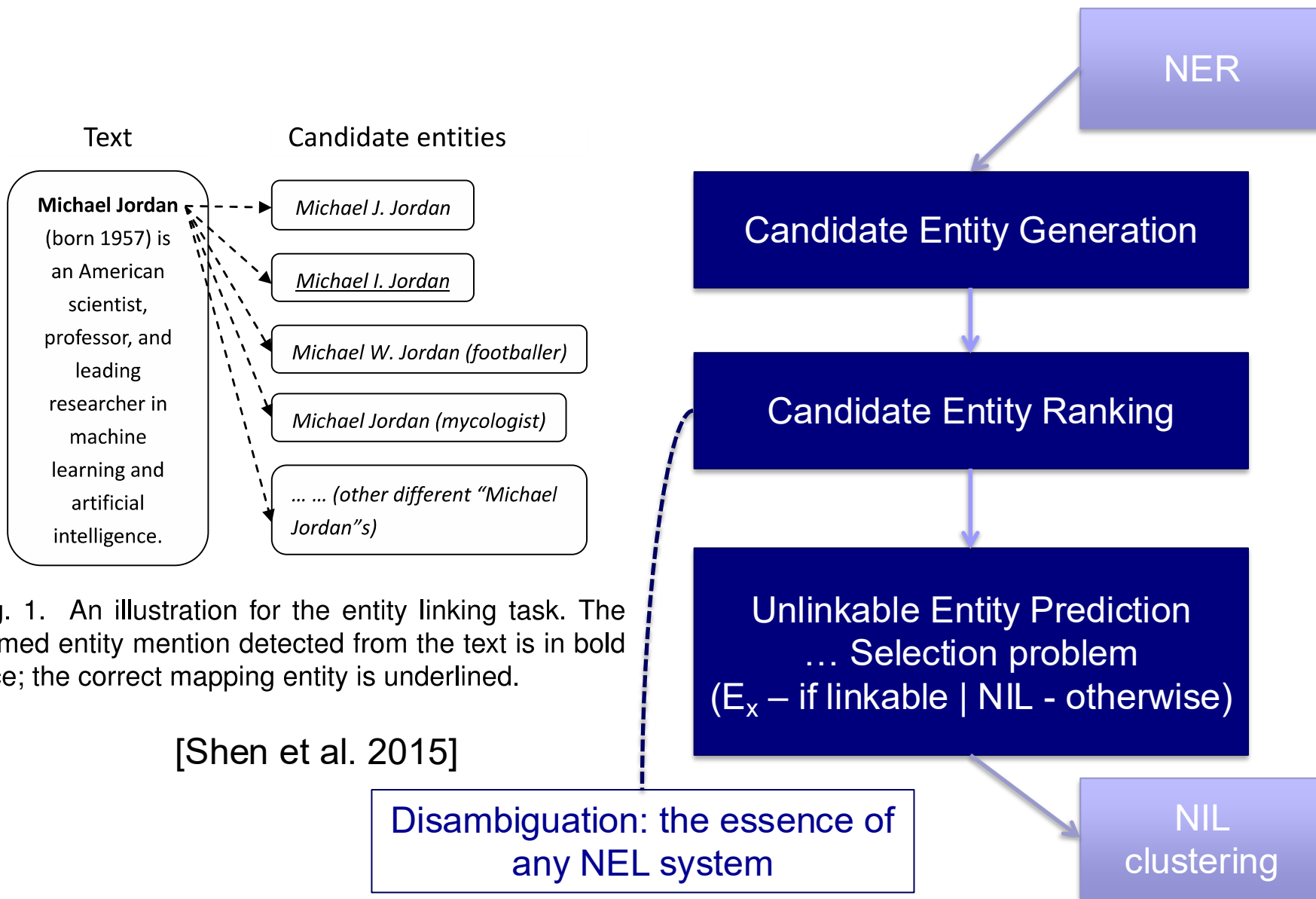


Fig. 1. An illustration for the entity linking task. The named entity mention detected from the text is in bold face; the correct mapping entity is underlined.

[Shen et al. 2015]

# NEL - Systems and approaches

## ■ NEL with Dbpedia/Wikipedia

- ☐ DBpedia Spotlight (<https://github.com/dbpedia-spotlight/>)
- ☐ GATE APIs (<https://gate.ac.uk/sale/tao/split.html>)
- ☐ TextRazor (<https://www.textrazor.com/>) - industrial
- ☐ Dandelion (<https://dandelion.eu/>) - industrial, IT
- ☐ Wikifier JSI (<http://wikifier.org/>) - multi-lingual
- ☐ Babelfy (<http://babelfy.org/>) - multi-lingual
  - Moro, Andrea, Alessandro Raganato, and Roberto Navigli. "Entity linking meets word sense disambiguation: a unified approach." Transactions of the Association for Computational Linguistics 2 (2014): 231-244.
- ☐ ...

# Simple Heuristic Linking Algorithm for Twitter

## ■ Candidate Resource Selection

- DBpedia articles (**rdfs:labels**) are indexed using Lucene
- Given an entity  $e_j$ , we compute a KB score w.r.t. each resource  $c_k$

$$KB(e_j, c_k) = \alpha \underbrace{lex(e_j, l_{c_k})}_{\text{lexical similarity}} + (1 - \alpha) \underbrace{cov(e_j, c_k)}_{\text{coverage}}$$

lexical similarity between an **entity**  $e_j$  and the **label** of a candidate resource  $l_{c_k}$

coverage based on the **coherence** of an entity  $e_j$  w.r.t. KB abstracts and the **popularity** of a candidate resource  $c_k$

# Pointers to Recent Neural Approaches

## ■ S-O-T-A 2018

- Kolitsas, N., Ganea, O. E., & Hofmann, T. (2018, October). **End-to-End Neural Entity Linking**. In Proceedings of the 22nd Conference on Computational Natural Language Learning (pp. 519-529).

- Papers with code link (obv. includes the code ☺ ):

- <https://paperswithcode.com/paper/end-to-end-neural-entity-linking>

## ■ Simple approach using BERT

- Broscheit, S. (2020). **Investigating Entity Knowledge in BERT with Simple Neural End-To-End Entity Linking**. Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL), 2019

- arXiv preprint arXiv:2003.05473.

## ■ Nice list of current topics and recent approaches:

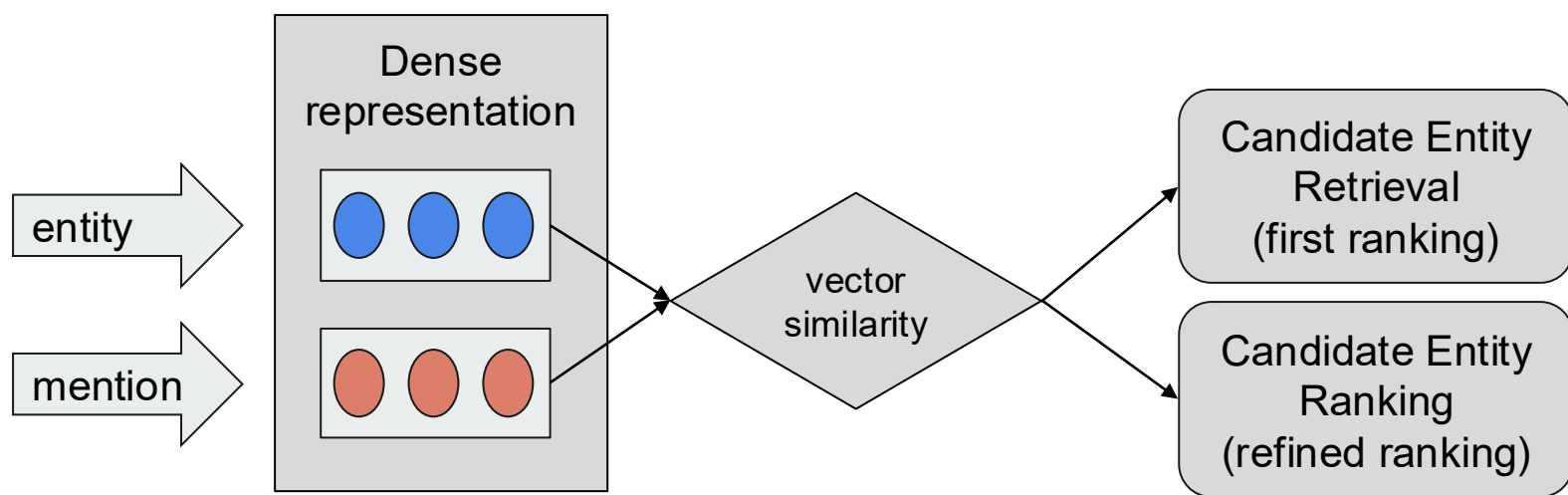
- [https://github.com/izuna385/EntityLinking\\_RecentTrend](https://github.com/izuna385/EntityLinking_RecentTrend)

## ■ Details on a recent approach later...



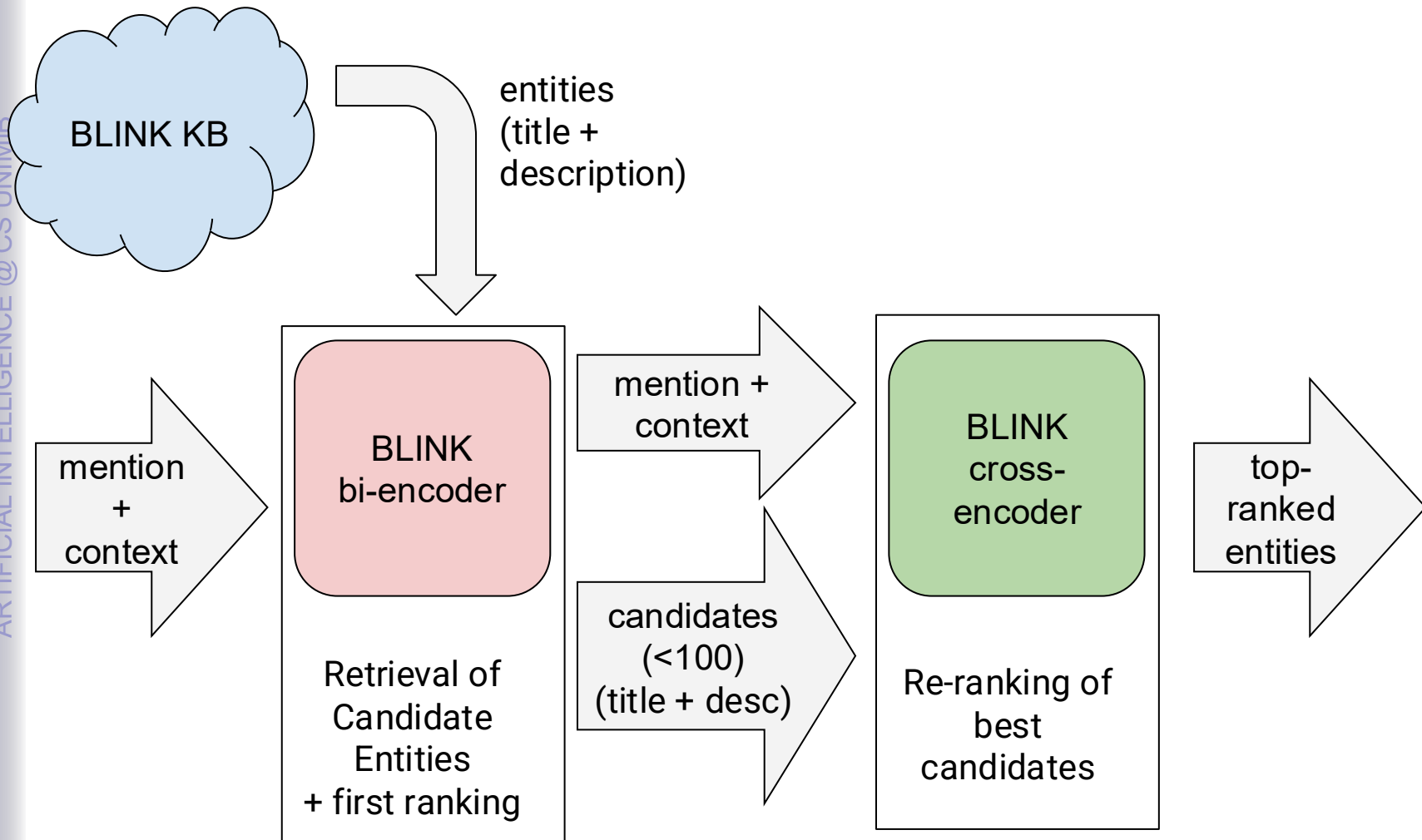
# NEL with Transformers: BLINK Idea

Recently **deep-learning** approaches obtained state-of-the-art results modeling mention context and entity descriptions.



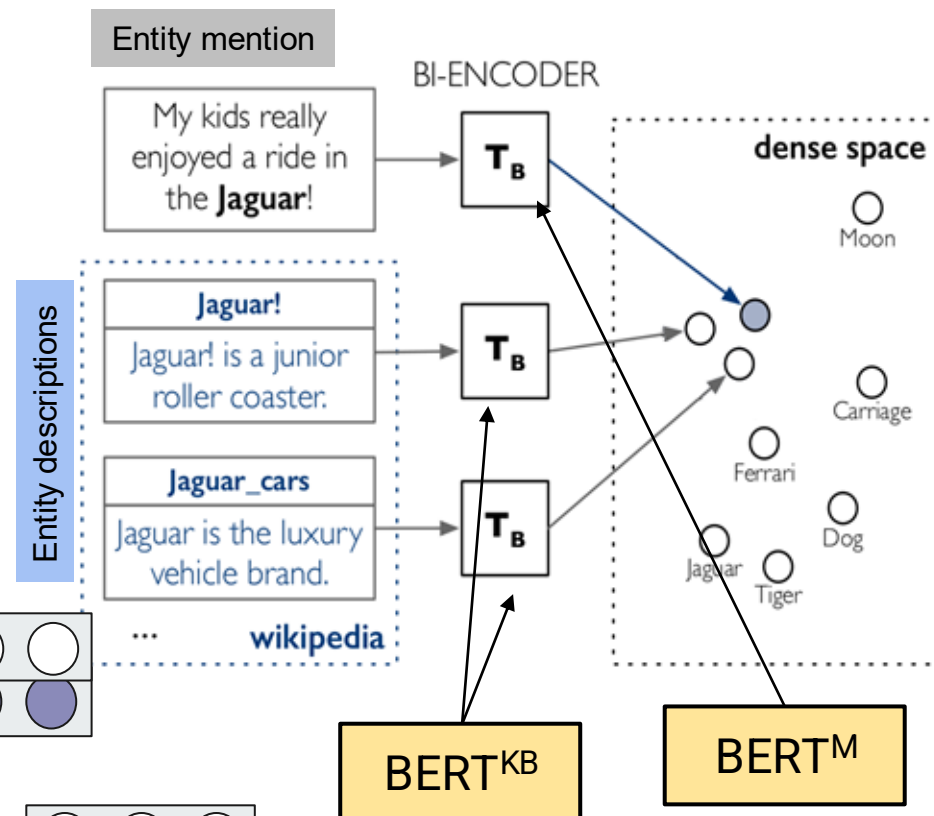
A more thoughtful approach based on BERT, particularly focused on efficient retrieval

# BLINK: Components



# BLINK: Bi-encoder Overview

- Based on BERT<sup>[2]</sup> \*.
- Idea:
  - fine tune BERT-based encoding of entity descriptions and mentions in context
  - in such a way they are close in the encoding space (alignment)
- Training:
  - Collect batch of mentions + linked entities
  - Get vectors:
    - title + description  $\rightarrow y_e$
    - mention + context  $\rightarrow y_m$
  - Compute similarity for pairs in  $Y_e \times Y_m$ :
    - dot\_prod( $y_e, y_m$ )
    - loss  $\rightarrow$  backprop
      - E.g., crossentropy

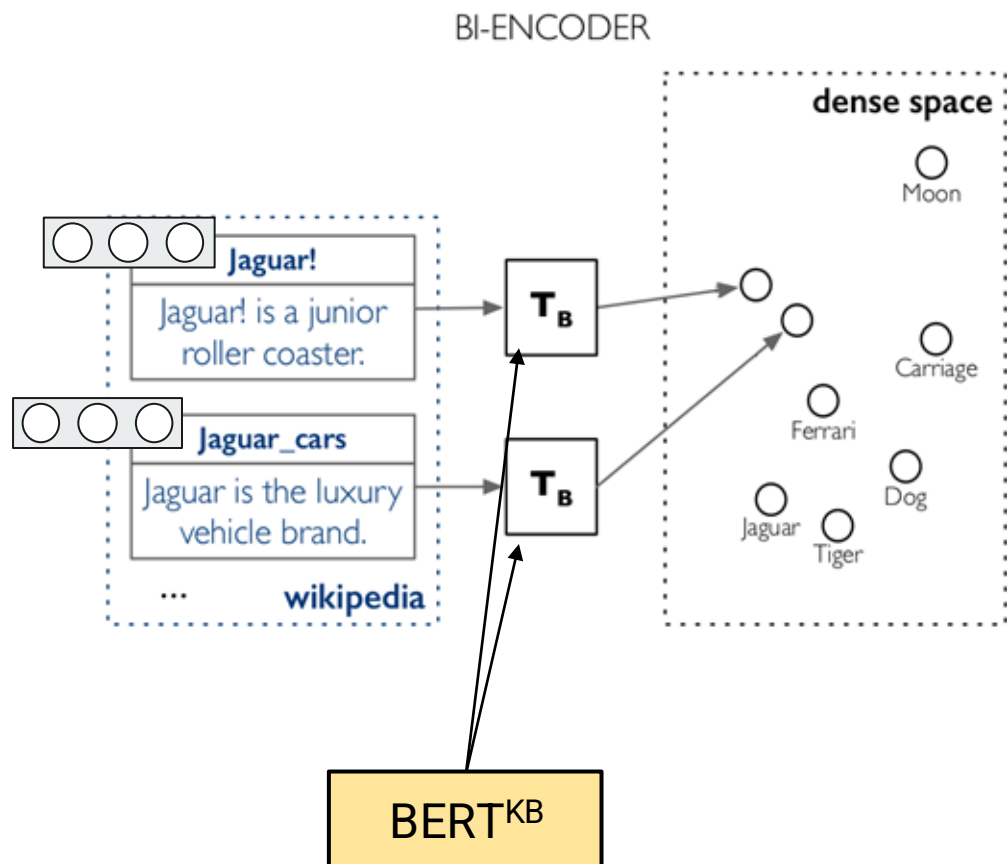


[1] Wu et al., 2020 - <https://doi.org/10.18653/v1/2020.emnlp-main.519>

[2] Devlin et al., 2018 - <http://arxiv.org/abs/1810.04805>

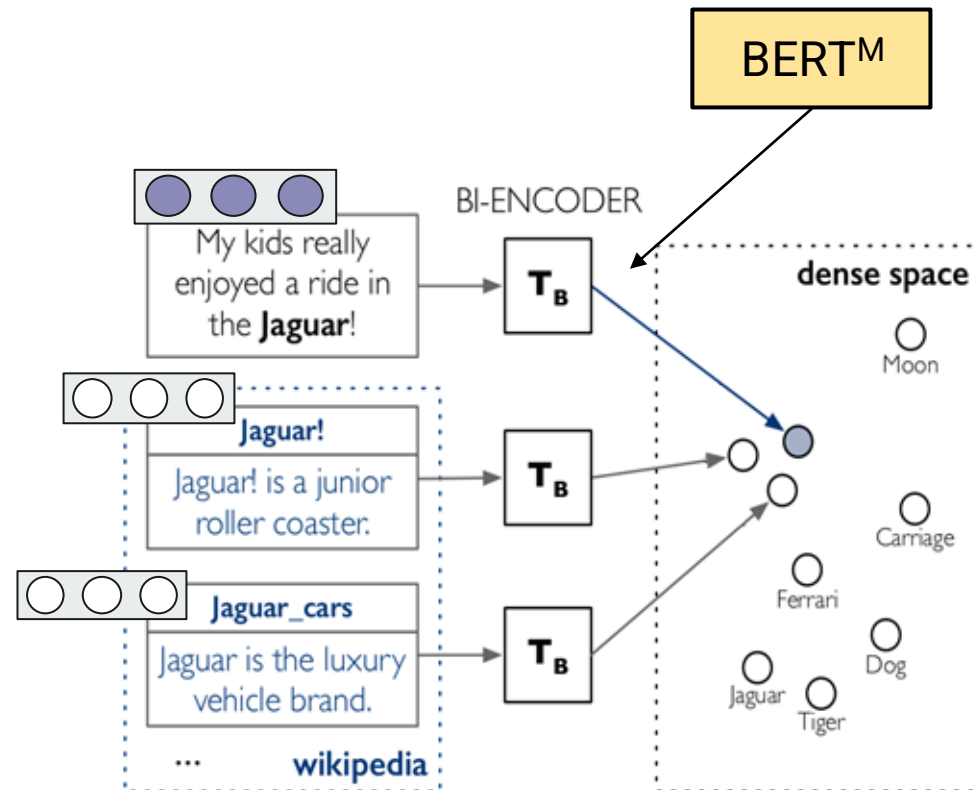
# NEL - BLINK - Bi-encoder / Entities

- BERT-based encoding:
  - title + description  $\rightarrow y_e$ 
    - [CLS] title [ENT] description [SEP]
    - Example of entity representation:
      - [CLS] *Jaguar!* [ENT] *Jaguard is a junior roller coaster* [SEP]
- The vector representation is from the **[CLS] token** of BERT's last layer.



# NEL - BLINK - Bi-encoder / Mentions

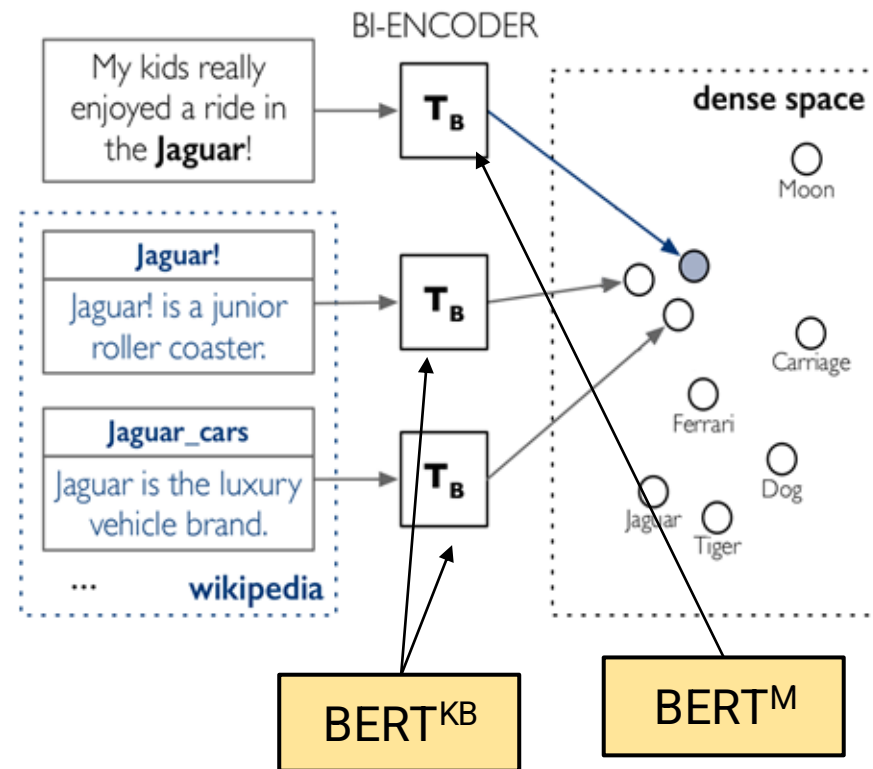
- BERT-based encoding:
  - title + description  $\rightarrow y_e$
  - mention+context  $\rightarrow y_m$ 
    - [CLS] *L-cont.* [MS] *mention*  
[ME] *R-cont.* [SEP]
  - Input example:
    - [CLS] *My...* [MS]  
*Jaguar* [ME] ! [SEP]
- Remark: the Bi-encoder NEVER receives both the mention and the entity together



# BLINK - Bi-encoder – Inference Time

(Preliminary) ranking based on vectors similarity (dot-product)

- INIT
  - [ Pre-compute entity vectors with fine-tuned BERT<sup>KB</sup> ]
- INFERENCE
  - Encode mention with BERT<sup>M</sup>
  - Candidate retrieval in the entity space with (approximate) **nearest neighbor**

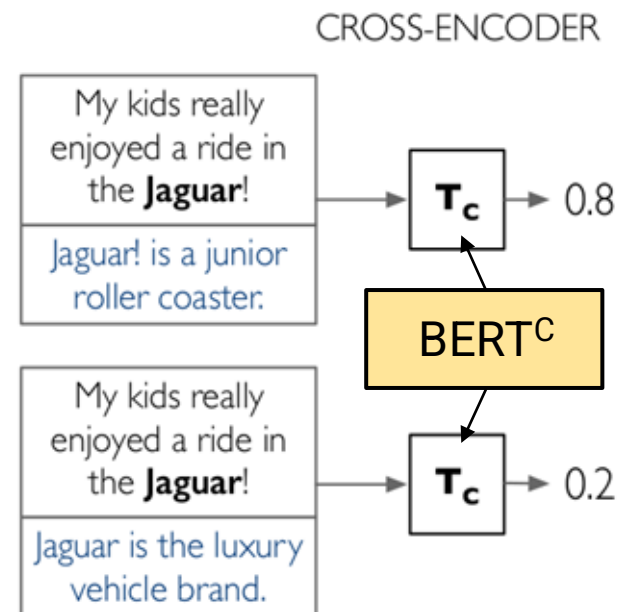


# NEL - BLINK - Cross-encoder

- Receives mention and entity **together** and determines a **matching score**
  - [CLS] **mention+cont.** [SEP] **title+description** [SEP]
    - **mention+cont.:** L-cont. [MS] mention [ME] R-cont.
    - **title+description:** title [ENT] description [SEP]
- Based on fine-tuning another BERT instance: BERT<sup>C</sup>
- Return a score for a [CLS] input
  - [CLS] → Linear layer → **score**.
- Applied to small set of candidate matching pairs
  - top-k (k=100) candidates (from the bi-encoder) for a mention

Used for Re-ranking (on a small set):

- more accurate
- but more expensive (memory + computation)



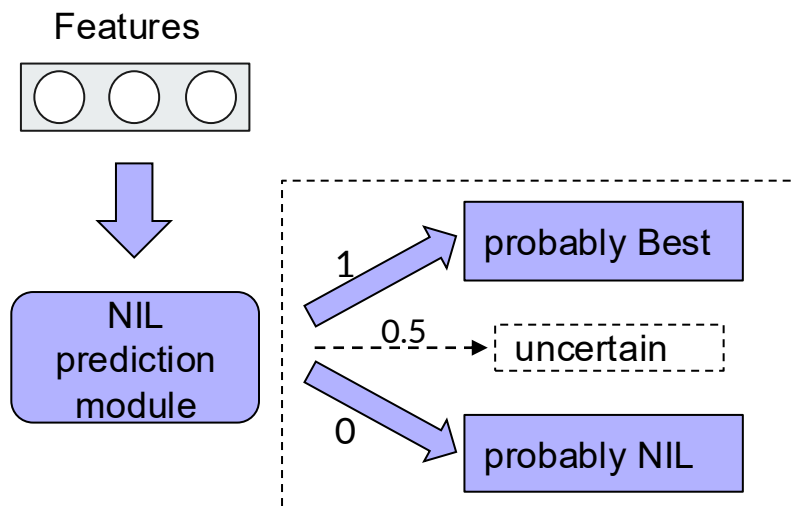
# NIL Prediction

Henry was successful in restoring power and stability to the English monarchy.

It is important to understand how confident the system is  
A human in some cases would answer “uncertain”, “don’t know”, or “probably X”

Not about a single candidate but about evidence for multiple candidates

- Estimate a **confidence score**  
interpretable by machine and  
**human validator**
- Logistic Regression model
  - with interpretable feature  
vectors
  - features: summary of results  
for top k candidates





# NIL Prediction: which Features?

*Henry was successful in restoring power and stability to the English monarchy.*

| Features                            | Overall Accuracy<br>(vs baseline) | Accuracy mitigated | 0-f1<br>NIL | 1-f1<br>not-NIL | macro-avg<br>f1<br>(vs max) |
|-------------------------------------|-----------------------------------|--------------------|-------------|-----------------|-----------------------------|
| <b>baseline*</b>                    | 66.5 (0.0)                        |                    |             |                 |                             |
| max (cross only)                    | 64.1 (-2.4)                       | 72.7               | 62.1        | 78.3            | 70.2 (-1.3)                 |
| max, jaccard                        | 65.6 (-0.9)                       | 73.9               | 63.0        | 79.4            | 71.2 (-0.3)                 |
| max, levenshtein                    | 65.8 (-0.7)                       | 74.0               | 63.1        | 79.6            | 71.4 (-0.1)                 |
| <b>max</b>                          | 65.8 (-0.7)                       | 74.2               | 63.2        | 79.8            | 71.5 (0.0)                  |
| max, stdev4                         | 65.3 (-1.2)                       | 75.2               | 66.5        | 80.0            | 73.2 (1.7)                  |
| max, stats10, levenshtein           | 66.2 (-0.3)                       | 76.3               | 67.9        | 81.0            | 74.4 (2.9)                  |
| max, jaccard, ner-types             | <b>69.7</b> (3.2)                 | 79.5               | 70.5        | 84.0            | 77.2 (5.7)                  |
| max, ner-types                      | <b>69.7</b> (3.2)                 | 79.7               | 70.6        | 84.1            | 77.4 (5.9)                  |
| max, stdev4, levenshtein, ner-types | 69.5 (3.0)                        | 79.9               | 71.4        | 84.1            | 77.8 (6.3)                  |
| <b>max, stats10, ner-types</b>      | <b>69.4</b> (2.9)                 | <b>80.0</b>        | <b>71.8</b> | <b>84.3</b>     | <b>78.0</b> (6.5)           |

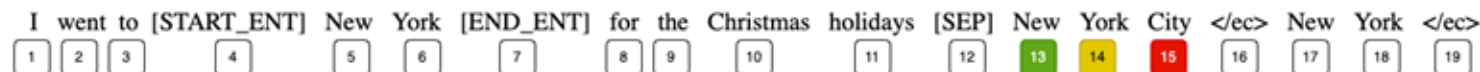
# Other Recent Approaches

## ■ Entity linking as extractive question answering [1]



The candidate list is added as context after [SEP]

The model makes a prediction on the **start** and **end** position of the correct candidate within the input set



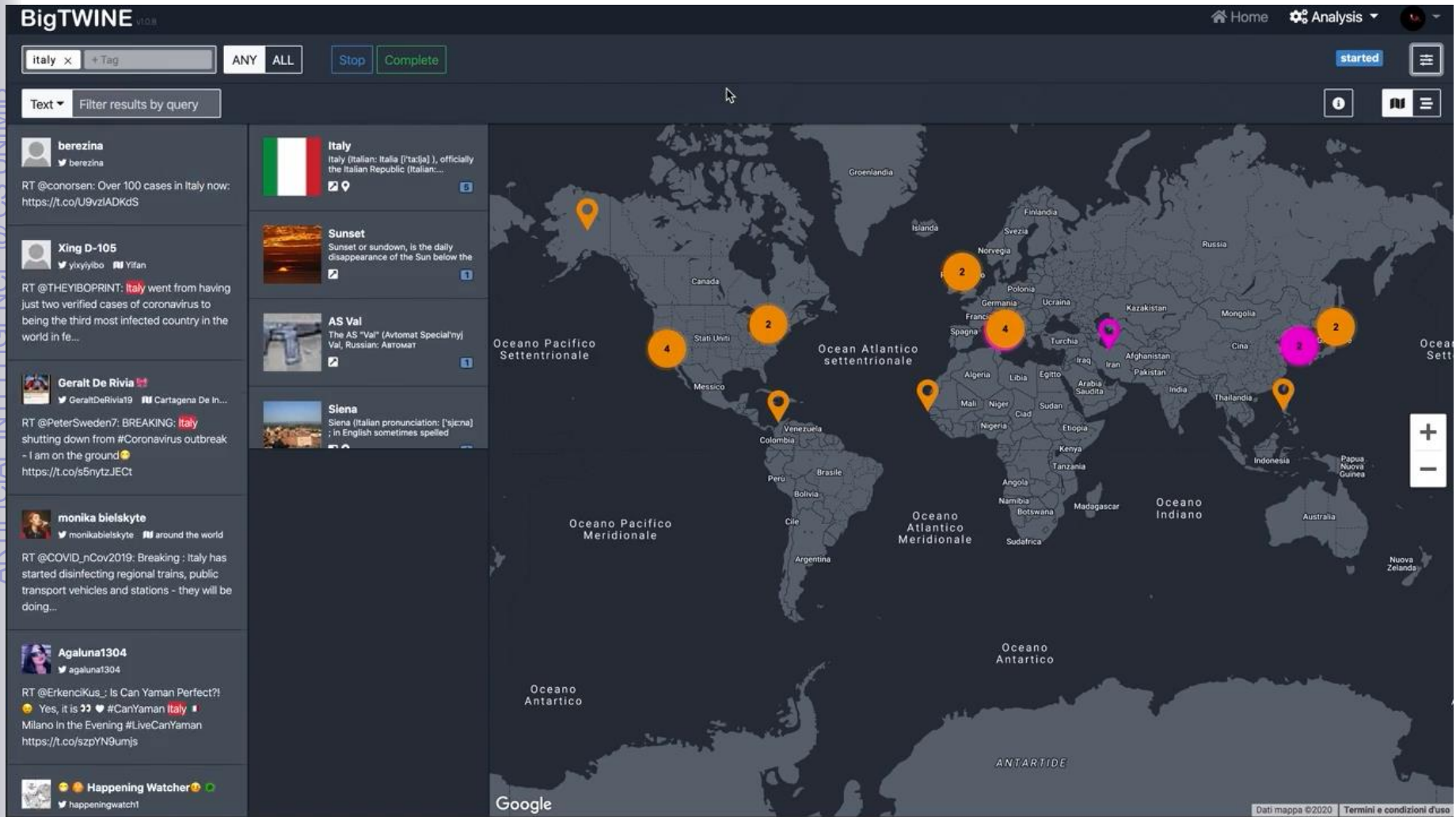
## ■ Generative Entity Linking [2]

- GENRE: generative approach with constrained decoding (only decodes tokens from a known set, i.e., entity ids, other special tokens)
- One follow-up work

[1] Barba, E., Procopio, L., & Navigli, R. (2022, May). ExtEnD: Extractive entity disambiguation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 2478-2488).

[2] De Cao, N., Izacard, G., Riedel, S., & Petroni, F. (2020, September). Autoregressive Entity Retrieval. In *ICLR 2021-9th International Conference on Learning Representations* (Vol. 2021). ICLR. <https://arxiv.org/abs/2010.00904>

# BigTWINE – BIG DATA PLATFORM FOR SEMANTIC ANALYSIS OF SOCIAL MEDIA



Check video at: <https://www.youtube.com/watch?v=xCJBunDtxMs&feature=youtu.be>

# NEL: Wrap-up

## ■ Named Entity Linking

- Candidate retrieval + re-ranking with link or not to link decision (disambiguation)
- In use today in industry and research
- Key ingredient for data enrichment: we can use links to fetch additional information, e.g., in the Twitter case

## ■ NEL with encoder-based LMs: BLINK

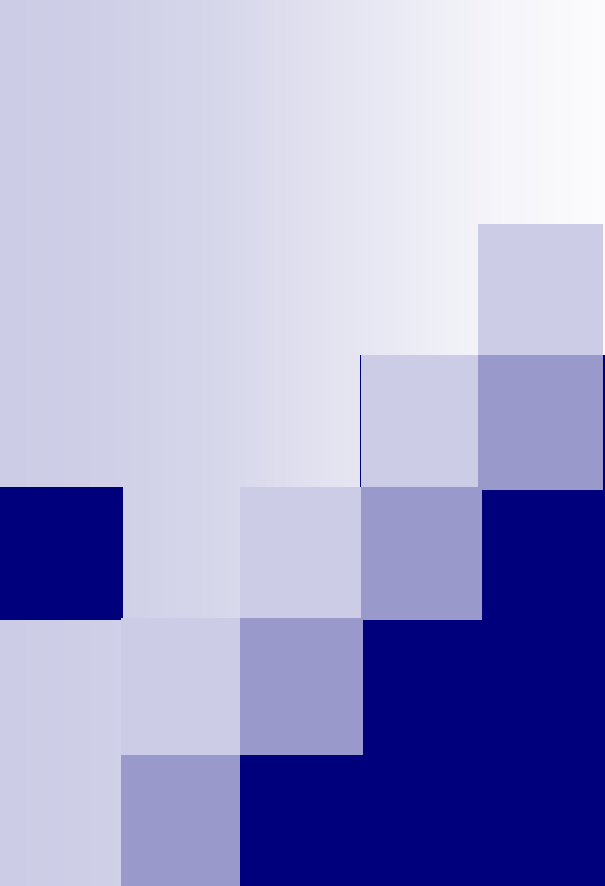
- Bi-encoder for entity retrieval: alignment of two encoders
- Cross-encoder for entity re-ranking: comparison based on sentence pair classification

## ■ Evolution and LLMs

- Speed up and avoid cross-encoding (not fast)
- Generative approaches exist, some example with tables later

# Outline

- Some preliminaries about linguistic analysis
- Named Entity Recognition
- Named Entity Linking
- **Relation Extraction**



# Relation Extraction

# Relation Extraction (RE)

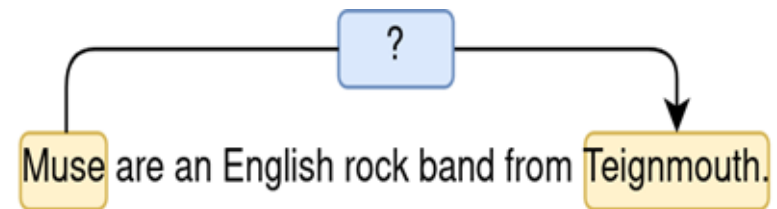
Relation Extraction (RE) is the task of extracting **semantic relationships between entities** within a text

Common assumption: we already know the entity mentions

Crucial step towards constructing large-scale KGs

RE flavours:

- multiclass vs. multilabel
- binary vs. n-ary
- focused vs. open



# Relation Extraction (RE)

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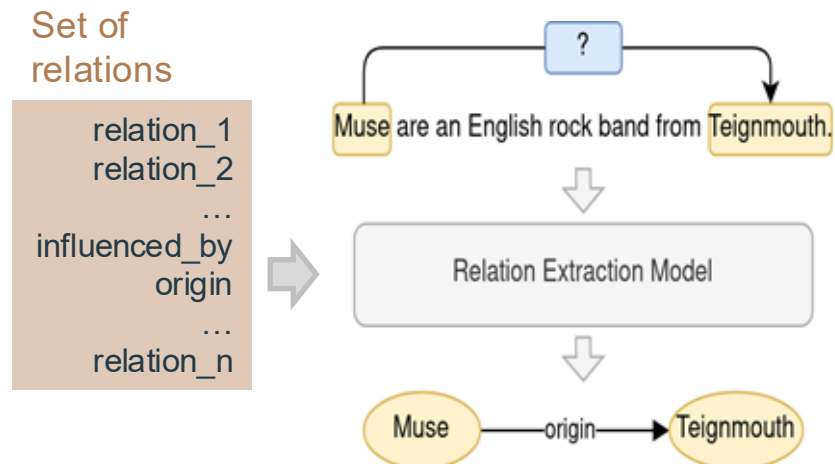
Crucial step towards constructing large-scale KGs

RE flavours:

- **multiclass** vs. multilabel
- **binary** vs. n-ary
- **focused** vs. open

Most common setting

- **Multiclass**: choose one relation from many
- **Binary**: as in KGs
  - e.g., `origin(Muse, Teignmouth)`
- **Focused**: consider only a pre-defined set of candidate relations





# Relation Extraction (RE)

Relation Extraction (RE) is the task of extracting **semantic relationships between entities** within a text

Common assumption: we already know the entity mentions

Crucial step towards constructing large-scale KGs

RE flavours:

- multiclass vs. multilabel
- binary vs. **n-ary**
- **focused** vs. open

Example of **n-ary relations** from GOLLIE

- Events
  - Fill in different “**slots**” for the relation, i.e., descriptive attributes

```
@atl class
class VulnerabilityPatch(Event):
    """A VulnerabilityPatch Event happens when a software company addresses a
    known vulnerability by releasing or describing an appropriate update."""

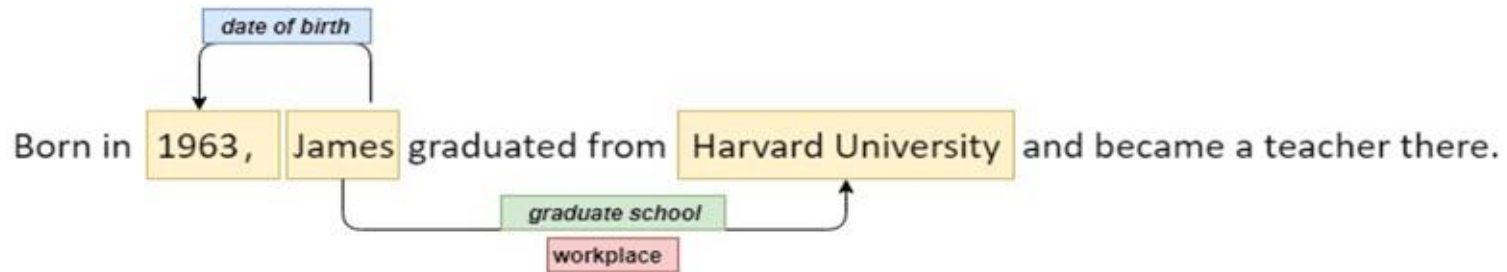
    mention: str
    """The text span that triggers the event.
    Such as: patch, fixed, addresses, implemented, released
    """

    cve: List[str] # The vulnerability identifier
    issues_addressed: List[str] # What did the patch fix
    supported_platform: List[str] # The platforms that support the patch
    vulnerability: List[str] # The vulnerability
    vulnerable_system: List[str] # The affected systems
    releaser: List[str] # The entity releasing the patch
    patch: List[str] # What was the patch about
    patch_number: List[str] # Number or name of the patch
    system_version: List[str] # The version of the vulnerable system
    time: List[str] # When was the patch implemented, the date
```

Slots to be filled in

... back to this example later 😊

# Relation Extraction Challenges: Examples for "influenced\_by"



- subject
- object
- influenced\_by implicit relation

... but remember: sentences can be of arbitrary complexity!  
Example: the **influenced\_by** relation (music domain)

- ex 1 | The Clash,<sup>[206]</sup> Siouxsie and the Banshees,<sup>[207]</sup> the Adverts,<sup>[208]</sup> Vic Godard of Subway Sect,<sup>[209]</sup> and Ari Up of the Slits<sup>[210]</sup> are among those in London's "inner circle" of early punk bands that credit the Pistols.
- ex 2 | Among the many musicians of a later time who have acknowledged their debt to the Pistols are members of the Jesus and Mary Chain,<sup>[211]</sup> NOFX,<sup>[212]</sup> The Stone Roses,<sup>[213]</sup> Guns N' Roses,<sup>[214]</sup> Nirvana,<sup>[215]</sup> Green Day,<sup>[200]</sup> and Oasis.<sup>[216]</sup> Calling the band "immensely influential", a London College of Music study guide notes that "many styles of popular music, such as grunge, indie, thrash metal and even rap owe their foundations to the legacy of ground breaking punk bands—of which the Sex Pistols was the most prominent."<sup>[1]</sup>
- ex 3 |

+ complexity

# Relation Extraction Techniques (1)

## Hand-crafted patterns (Hearst, 1992): \*

- high precision
- low recall (poor generalization)
- lot of work to think of all possible patterns
- don't want to do this for every relation

| Hearst pattern  | Example occurrences                                          |
|-----------------|--------------------------------------------------------------|
| X and other Y   | ...temples, treasures, and other important civic buildings.  |
| X or other Y    | bruises, wounds, broken bones or other injuries...           |
| Y such as X     | The bow lute, such as the Bambara ndang...                   |
| such Y as X     | ...such authors as Herrick, Goldsmith, and Shakespeare.      |
| Y including X   | ...common-law countries, including Canada and England...     |
| Y, especially X | European countries, especially France, England, and Spain... |

{'LEMMA': {'IN': ['cite', 'refer', 'list', 'mention', 'credit', 'claim']}},  
"POS": 'VERB'},  
{'OP': ""},  
{'LEMMA': 'be'},  
{'OP': ""},  
{'LEMMA': 'influence', "POS": 'NOUN'}

... matches, e.g.,:

*"Michael Jackson claimed that James Brown was his main influence"*

*The text pattern matches all sentences that contain (in the specified order)*

- A verb among {cite, refer, ...claim}
- An arbitrary sequence of tokens
- The verb {be}
- An arbitrary sequence of tokens
- The noun {influence}

# Relation Extraction Techniques (1)

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| such Y as X     | ...such authors as Herrick, Goldsmith, and Shakespeare.      |
| Y including X   | ...common-law countries, including Canada and England...     |
| Y, especially X | European countries, especially France, England, and Spain... |

{'LEMMA': {'IN': ['cite', 'refer', 'list', 'mention', 'credit', 'claim']}},  
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... matches, e.g.,:

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You can check also rule-based  
matchers in spaCy

<https://spacy.io/usage/rule-based-matching>

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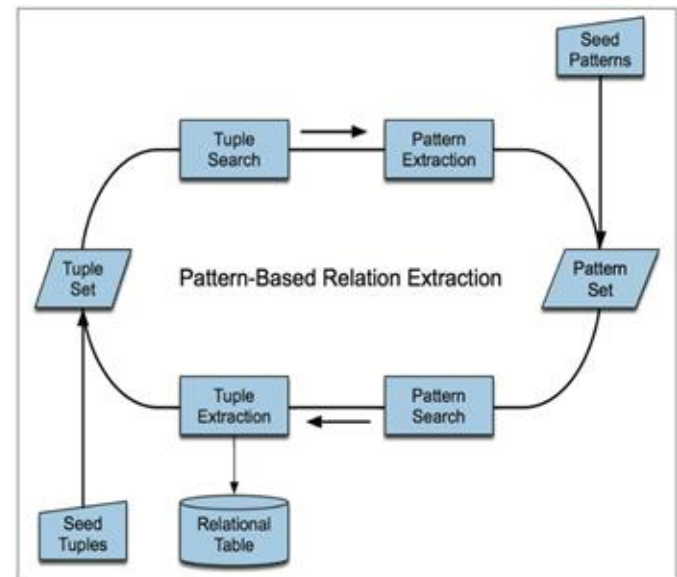
| Hearst pattern  | Example occurrences                                          |
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| Y such as X     | The bow lute, such as the Bambara ndang...                   |
| such Y as X     | ...such authors as Herrick, Goldsmith, and Shakespeare.      |
| Y including X   | ...common-law countries, including Canada and England...     |
| Y, especially X | European countries, especially France, England, and Spain... |

## Bootstrapping methods (Brin, 1992):

starting from seed instances and patterns this virtuous circle find new patterns and instances in large amounts of text

- better recall
- semantic drift

SKIP



# Relation Extraction Techniques (2)

SKIP

## Supervised:

- not limited to one relation per model
- uses features like POS, NER, dependency tree, etc
- can get high accuracies with enough hand-labeled training data
- now based on DNNs (e.g., with word embeddings)
- build the training set is expensive

SKIP

## Distant Supervised (Mintz, 2009): \*

Use a KB instead of a set of labeled data

- positives samples → for each instance of the relation in the KB, select sentences that match these tuples.
- negative samples → select sentences with couples of entities where the relation doesn't hold on the KB.

Featurize and train a classifier

- less manual effort but strong assumption
- we cannot learn relations that are not in the KB

# BERT-based RE Example

Wu, S., & He, Y. **Enriching pre-trained language model with entity information for relation classification**. In CIKM 2019. ACM (pp. 2361-2364).

Arxiv version: <https://arxiv.org/abs/1905.08284> // Code:

<https://github.com/monologg/R-BERT>

## Relations:

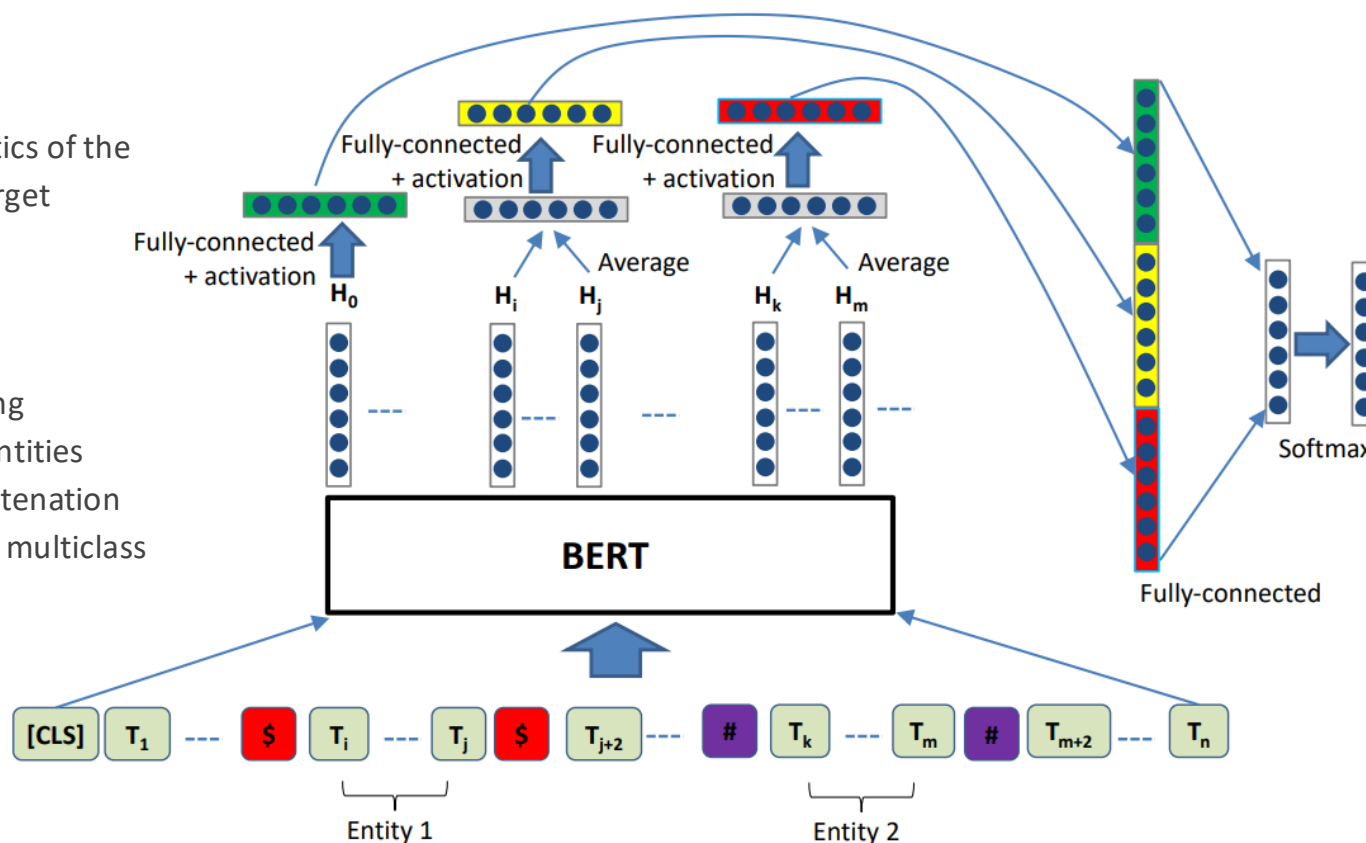
- Cause-Effect
- Component-Whole
- Content-Container
- EntityDestination
- Entity-Origin
- Instrument-Agency
- Member-Collection
- Message-Topic
- ProductProduce

intuition:

- captures semantics of the sentence and target entities

model details:

- BERT for encoding sentences and entities
- encodings concatenation through relation multiclass classifier



# Relation Extraction with Gen LLMs

- Naïve prompting is not enough
- Careful prompting and good examples work quite well

|                                                  | Methods                                           | Retriever | Semeval    | TACRED     | SciERC     | ACE05     |
|--------------------------------------------------|---------------------------------------------------|-----------|------------|------------|------------|-----------|
| Random examples: poor<br>(GPT4 is likely better) | <i>GPT-3 Baselines (Best <math>k</math>-shot)</i> |           |            |            |            |           |
|                                                  | GPT-Random                                        | -         | 70.04 (30) | 32.49 (15) | 17.92 (25) | 9.04 (25) |
|                                                  | GPT-Sent                                          | SimCSE    | 79.94 (30) | 33.45 (15) | 20.96 (25) | 6.31 (25) |

|                  |                                |  |              |        |       |              |
|------------------|--------------------------------|--|--------------|--------|-------|--------------|
| State-of-the-art | <i>Fine-tuned RE Baselines</i> |  |              |        |       |              |
|                  | Cohen et al. (2020)            |  | <b>91.90</b> | -      | -     | -            |
|                  | Wang et al. (2022a)            |  | -            | ♣76.80 | -     | -            |
|                  | PURE (Zhong and Chen, 2021)    |  | 89.90        | 69.72  | 68.45 | <b>70.09</b> |

Table 2: **Main Results on four RE datasets.** All results are given by Micro-F1. \* denotes the same  $k$ -shot for the comparison with + Reasoning. Due to the costly GPT-3 expense, we conducted Reasoning experiments on the two relatively smaller datasets Semeval and TACRED. ♣ denotes that this performance is not comparable as it evaluates on the entire test set. The underline denotes the results outperforming the fine-tuning baseline PURE.



# Relation Extraction with Gen LLMs

- Naïve prompting is not enough
- **Careful prompting** and good examples work quite well

Event extraction:  
n-ary relation  
extraction task in  
GOLLIE

Special prompt  
engineering  
+ examples (ICL)  
+ fine-tuning

```
@dataclass
class VulnerabilityPatch(Event):
    """A VulnerabilityPatch Event happens when a software company addresses a
    known vulnerability by releasing or describing an appropriate update."""

    mention: str
    """The text span that triggers the event.
    Such as: patch, fixed, addresses, implemented, released
    """

    cve: List[str] # The vulnerability identifier
    issues_addressed: List[str] # What did the patch fix
    supported_platform: List[str] # The platforms that support the patch
    vulnerability: List[str] # The vulnerability
    vulnerable_system: List[str] # The affected systems
    releaser: List[str] # The entity releasing the patch
    patch: List[str] # What was the patch about
    patch_number: List[str] # Number or name of the patch
    system_version: List[str] # The version of the vulnerable system
    time: List[str] # When was the patch implemented, the date
```

# Relation Extraction with Gen LLMs

- Naïve prompting is not enough
- Careful prompting and **good examples** work quite well

Dedicated approach  
fine-tuned to retrieve  
good examples  
(BERT-based)

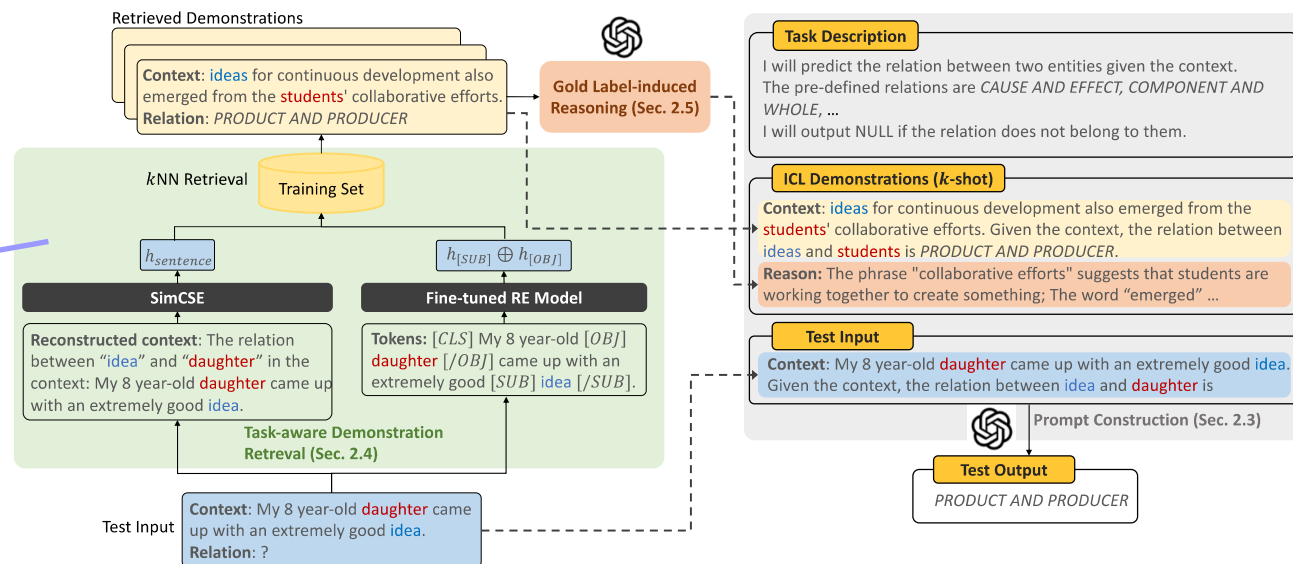


Figure 4: **An illustration of GPT-RE.** Given a test input, we first leverage two different task-aware retrieval methods to search for highly relevant demonstrations from the training set, and then incorporate the gold label-induced reasoning for each demonstration. Above contents will then be included in the prompt construction to make the prediction.

# Relation Extraction with Gen LLMs

- Naïve prompting is not enough
- Careful prompting and **good examples** work quite well

|                                                              | Methods                              | Retriever | Semeval           | TACRED            | SciERC            | ACE05        |
|--------------------------------------------------------------|--------------------------------------|-----------|-------------------|-------------------|-------------------|--------------|
| Random examples: poor<br>(GPT4 is likely better)             | <i>GPT-3 Baselines (Best k-shot)</i> |           |                   |                   |                   |              |
|                                                              | GPT-Random                           | -         | 70.04 (30)        | 32.49 (15)        | 17.92 (25)        | 9.04 (25)    |
|                                                              | GPT-Sent                             | SimCSE    | 79.94 (30)        | 33.45 (15)        | 20.96 (25)        | 6.31 (25)    |
| Proposed approach<br>... with examples<br>... FT / Reasoning | <i>Ours (Best k-shot)</i>            |           |                   |                   |                   |              |
|                                                              | GPT-RE_SimCSE                        | SimCSE    | 81.02 (30)        | 37.44 (15)        | 26.46 (25)        | 8.67 (25)    |
|                                                              | GPT-RE_SimCSE*                       | SimCSE    | 77.49 (15)        | 31.58 (10)        | -                 | -            |
|                                                              | + Reasoning                          | SimCSE    | 79.88 (15)        | 33.18 (10)        | -                 | -            |
|                                                              | GPT-RE_FT                            | PURE      | <u>91.90</u> (25) | <u>72.14</u> (15) | <u>69.00</u> (30) | 68.73 (25)   |
|                                                              | GPT-RE_FT*                           | PURE      | <u>91.11</u> (15) | <u>70.38</u> (10) | -                 | -            |
| State-of-the-art                                             | <i>Fine-tuned RE Baselines</i>       |           |                   |                   |                   |              |
|                                                              | Cohen et al. (2020)                  |           | <b>91.90</b>      | -                 | -                 | -            |
|                                                              | Wang et al. (2022a)                  |           | -                 | ♣ <b>76.80</b>    | -                 | -            |
|                                                              | PURE (Zhong and Chen, 2021)          |           | 89.90             | 69.72             | 68.45             | <b>70.09</b> |

Table 2: **Main Results on four RE datasets.** All results are given by Micro-F1. \* denotes the same  $k$ -shot for the comparison with + Reasoning. Due to the costly GPT-3 expense, we conducted Reasoning experiments on the two relatively smaller datasets Semeval and TACRED. ♣ denotes that this performance is not comparable as it evaluates on the entire test set. The underline denotes the results outperforming the fine-tuning baseline PURE.

# IE Wrap-up

- Terminology and task definition
- Examples of specific approaches (rule-based vs. neural)
  - NER / NEL + NIL Prediction / RE
  - LLM x Information Extraction
- Some additional topics
  - Fine-grained classification (entities)
  - Generation of synthetic data with NLG
  - Adaptation
- Maturity levels
  - LLM may change maturity, e.g., for RE (costs!)
  - Remember... batch processing may require efficiency

# LMs & Information Extraction

Fundamental bricks for modern IE

- Embeddings and encoders: signals to classifiers
  - Tagging
  - Bi-encoders and cross-encoders for matching-related tasks
- LLMs and generative models: return output
  - Prompt engineering
  - Single-turn vs. multi-turn prompts
  - Demonstrations and ICL

# LLMs & Information Extraction

LLMs (vs. other approaches exploiting transformers)

## ■ Pros:

- Benefit from world knowledge, language skills, chains-of-thoughts and reasoning (and dedicated FT & alignment strategies?)
- Especially promising in hard tasks, e.g., RE
- Powerful in structuring information, e.g., tables

## ■ Cons (today):

- Non-native support for positional annotations
- Black-box approaches: lack of insights on uncertainty
- Data sharing & privacy (top-tier API-based models)
- Fine-tuning and performance (smaller on-premise models)
- Costs and speed, esp. for more sophisticated patterns